MA 232 Course Project 2:
How AI Detects Objects: A Linear Algebra Approach to Pattern Recognition

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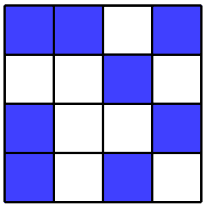
"I pledge my honor that I have abided by the Stevens Honor System."

Part A: Convert the Pattern into a Vector

Task A.1: Target Grid and Vector Representation

Convert the given grid into a vector $w \in \mathbb{R}^{16}$.

Answer: The target pattern is a 4×4 binary grid with blue cells as 1 and white cells as 0:



Reading row by row from top to bottom (left to right) yields the target vector $w \in \mathbb{R}^{16}$:

$$w = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}^T$$

Task A.2: Cell Counts

How many blue cells and white cells does the target pattern contain?

Answer: The target pattern contains **8 blue cells** (value 1) and **8 white cells** (value 0), for a total of 16 cells.

Task A.3: General Pattern Encoding Procedure

Explain how an arbitrary blue-and-white 4×4 grid can be represented by a vector $u \in \{0, 1\}^{16}$.

Answer: We can treat the 4×4 grid like a coordinate system where each cell has a row i and column j (from 1 to 4). Each cell takes a value of 1 if it is filled with blue, or 0 if it is white.

To flatten this into a vector, we read the grid row by row from top to bottom (left to right) into a 16-element column vector $u \in \{0,1\}^{16}$. Because there are 16 cells and each cell can be either 0 or 1, this vector representation accounts for all $2^{16} = 65,536$ possible blue-and-white patterns.

Part B: Construct a Matrix for Pattern Recognition

Task B.1: Information Represented by Vector u

Explain what information is represented by the vector u.

Answer: The vector $u \in \{0,1\}^{16}$ represents any 4×4 test pattern we want to check against our target pattern. Each of the 16 entries in u tells us whether that specific box on the grid is blue (1) or white (0).

Task B.2: Role of Quadratic Form $Q(u) = u^T Mu$

Describe the role of the quadratic form $Q(u) = u^T Mu$ in recognizing a pattern.

Answer: The quadratic form $Q(u) = u^T Mu$ works like a scoring function for any test pattern u:

- If the function returns 0, the test pattern matches our target pattern ($Q(w) = w^T Mw = 0$).
- If the function returns a nonzero (positive) score, the pattern is not the target pattern ($Q(u) \neq 0$).

This turns pattern recognition into a simple check: an input pattern is recognized as the target pattern if and only if $Q(u) = 0$.

Task B.3: Construction of Matrix M

Construct a 16×16 matrix M that satisfies the two conditions: $w^T Mw = 0$ and $u^T Mu \neq 0$ for every other nonzero binary vector $u \neq w$.

Answer: We construct M in the form $M = B^T B$. To guarantee that $Mw = \mathbf{0}$, we choose the rows of B to be 15 vectors $v_1, \ldots, v_{15}$ that are all perpendicular (orthogonal) to w.

1. Setting the dot product $w \cdot v = 0$ using our target vector gives:

$$v_1 + v_2 + v_4 + v_7 + v_9 + v_{12} + v_{13} + v_{15} = 0$$

2. Solving for the first component:

$$v_1 = -(v_2 + v_4 + v_7 + v_9 + v_{12} + v_{13} + v_{15})$$

   Setting each of the other 15 free components to 1 (one at a time, with all others 0) produces a basis of 15 vectors $\{v_1, v_2, \ldots, v_{15}\}$ that are all orthogonal to w.

3. We stack these 15 vectors as rows of a 15×16 matrix B:

$$B = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_{15}^T \end{bmatrix}$$

4. We set $M = B^T B$, which produces our 16×16 matrix.

Task B.4: Verification of the Null Target Condition

Show that your matrix satisfies $w^T M w = 0$.

Answer: Because every row of B was chosen to be orthogonal to w, multiplying B by w gives:

$$Bw = \begin{bmatrix} v_1^T w \\ \vdots \\ v_{15}^T w \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}$$

This means $Mw = B^T(Bw) = B^T\mathbf{0} = \mathbf{0}$.

Evaluating the quadratic form for the target vector gives:

$$w^T M w = w^T(Mw) = w^T\mathbf{0} = 0$$

Task B.5: Nonzero Condition for Non-Target Binary Vectors

Explain why the quadratic form must be nonzero for every other nonzero binary vector.

Answer: Using the transpose product rule, the quadratic form can be written as:

$$u^T M u = u^T(B^T B)u = (Bu)^T(Bu) = \|Bu\|^2$$

Because $\|Bu\|^2$ is a squared length, it is always ≥ 0, and equals 0 if and only if $Bu = \mathbf{0}$. The only vectors satisfying $Bu = \mathbf{0}$ are scalar multiples of the target pattern, $u = c \cdot w$.

Since u is restricted to binary patterns (0 and 1 only):

- $c = 0 \implies u = \mathbf{0}$ (the all-white grid).
- $c = 1 \implies u = w$ (the target pattern).

For any other nonzero binary vector $u \neq w$, u is not a multiple of w, which means $Bu \neq \mathbf{0}$. This guarantees that the score is strictly positive:

$$u^T M u = \|Bu\|^2 > 0 \neq 0$$

Task B.6: Symmetry and Invertibility of M

State whether your matrix is symmetric and whether it is invertible. Justify your answers.

Answer:

- **Symmetry: Yes.** Using the transpose product rule:

$$M^T = (B^T B)^T = B^T (B^T)^T = B^T B = M$$

  Since $M^T = M$, matrix M is symmetric.

- **Invertibility: No.** From Task B.4, we know that $Mw = \mathbf{0}$. Because matrix M maps a nonzero vector ($w \neq \mathbf{0}$) to the zero vector, w is a non-trivial vector in the nullspace of M. Therefore, $\det(M) = 0$, its rank is 15 (not full rank 16), and M is not invertible.

Part C: Verify the Recognition Rule

Task C.1: Systematic Verification Methodology

Develop a systematic method to verify your matrix.

Answer: To verify our matrix M, we first check that the target pattern produces $Q(w) = 0$. We then test other patterns to make sure they produce nonzero values. Finally, we use Python to check all 65,535 possible nonzero binary patterns and confirm that only the target produces zero.

Task C.2: Verification of the Target Vector

Verify the quadratic-form value for the target vector.

Answer: To verify that the target pattern produces zero, we use the matrix constructed in Part B, where $M = B^T B$. Substituting this into the quadratic form gives:

$$Q(w) = w^T M w = w^T B^T B w$$

We can rewrite this as:

$$Q(w) = (Bw)^T (Bw)$$

Since B was constructed so that $Bw = 0$, we get:

$$Q(w) = 0^T 0 = 0$$

Therefore, the target vector satisfies the recognition condition $w^T M w = 0$.

Task C.3: Testing on Required Binary Patterns

Test your matrix on several other binary patterns, including: a grid with only one blue cell, a grid with two blue cells, a grid with three blue cells in different positions, and the grid with all 16 cells blue.

Answer: We tested our matrix M using several different binary patterns and calculated $Q(u) = u^T M u$ for each.

Test Pattern	$Q(u)$
One blue cell (Cell 1)	7
One blue cell (Cell 3)	1
Two blue cells (Cells 1, 2)	6
Two blue cells (Cells 1, 3)	8
Three blue cells (Cells 1, 2, 4)	5
Three blue cells (Cells 3, 5, 6)	3
All 16 cells blue	8

All tested patterns produced nonzero values. This confirms that our matrix correctly rejects these patterns because none matches the target vector w.

Task C.4: Exhaustive Verification of All $2^{16} - 1 = 65,535$ Nonzero Binary Vectors

Explain how you can verify the condition for all $2^{16} - 1 = 65,535$ nonzero binary vectors.

Answer: There are $2^{16} = 65,536$ possible binary vectors for a 4×4 grid. Excluding the all-zero vector leaves 65,535 nonzero patterns.

We can use Python to generate every possible binary vector and calculate $Q(u) = u^T M u$ for each one. We then check that the target vector w is the only nonzero vector that produces zero.

This method allows us to verify every possible pattern rather than relying only on a few examples.

Task C.5: Table of Test Patterns and Values

Present a table of at least 10 test patterns, their binary vectors, and the corresponding quadratic-form values.

Table 1 presents the quadratic-form values calculated using matrix M for 10 different binary patterns.

#	Test Pattern	Binary Vector	$Q(u)$
1	Target	1101001010011010	0
2	Cell 1	1000000000000000	7
3	Cell 3	0010000000000000	1
4	Cells 1, 2	1100000000000000	6
5	Cells 1, 3	1010000000000000	8
6	Cells 1, 2, 4	1101000000000000	5
7	Cells 3, 5, 6	0010110000000000	3
8	Cells 1, 2, 4, 7	1101001000000000	4
9	Inverted target	0010110101100101	8
10	All blue	1111111111111111	8

Table 1: Test patterns and quadratic-form values.

Task C.6: Generality Beyond the Sample Table

State how you know that your construction satisfies the requirement for all nonzero binary patterns, not merely the patterns in your sample table.

Answer: Our matrix M was constructed as $M = B^T B$, where the null space of B is spanned by the target vector w.

Therefore, $Q(u) = u^T M u = (Bu)^T (Bu) = 0$ only when u is a multiple of w.

Since we are only considering binary vectors, the only possible multiples are the zero vector and w itself. The zero vector is excluded from the requirements.

Therefore, w is the only nonzero binary vector that produces zero, proving that our matrix works for all 65,535 nonzero patterns.

Part D: Explore Different Matrix Constructions

D.1: Construct a Second Matrix

Task D.1.a: Construction of Second Matrix N

Find a second 16×16 *matrix* N*, different from your original matrix* M*, that recognizes the same target pattern.*

Answer: We construct a second matrix N using the formula

$$N = 8I - ww^T$$

where I is the 16×16 identity matrix and w is our target vector.

Since w contains eight ones, $w^T w = 8$. This gives us a different matrix from M that recognizes the same target pattern.

$$N = \begin{bmatrix}
7 & -1 & 0 & -1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & -1 & -1 & 0 & -1 & 0 \\
-1 & 7 & 0 & -1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & -1 & -1 & 0 & -1 & 0 \\
0 & 0 & 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & -1 & 0 & 7 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & -1 & -1 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & -1 & 0 & -1 & 0 & 0 & 7 & 0 & -1 & 0 & 0 & -1 & -1 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & -1 & 0 & -1 & 0 & 0 & -1 & 0 & 7 & 0 & 0 & -1 & -1 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 8 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 8 & 0 & 0 & 0 & 0 & 0 \\
-1 & -1 & 0 & -1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 7 & -1 & 0 & -1 & 0 \\
-1 & -1 & 0 & -1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & -1 & 7 & 0 & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 8 & 0 & 0 \\
-1 & -1 & 0 & -1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & -1 & -1 & 0 & 7 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 8
\end{bmatrix}$$

Task D.1.b: Verification of Conditions for Matrix N

Verify that $w^T N w = 0$ and $u^T N u \neq 0$ for every other nonzero binary vector $u \neq w$.

Answer: To verify that N recognizes the target, we calculate:

$$w^T N w = w^T (8I - ww^T) w$$

$$= 8(w^T w) - (w^T w)^2$$

$$= 8(8) - 8^2 = 64 - 64 = 0$$

For any vector u, we have:

$$u^T N u = 8\|u\|^2 - (w^T u)^2$$

By the Cauchy-Schwarz inequality,

$$(w^T u)^2 \leq 8\|u\|^2$$

Equality occurs only when u is a multiple of w. Since u is binary, the only possibilities are $u = 0$ and $u = w$.

Therefore, every other nonzero binary vector produces a positive value, so N satisfies the recognition conditions.

Task D.1.c: Comparison Between Matrices M and N

Explain how the second matrix differs from the first and why both matrices work.

Answer: Our original matrix $M = B^T B$ was constructed using vectors perpendicular to the target w.

Our second matrix $N = 8I - ww^T$ was constructed using an orthogonal projection method.

Although the matrices contain different values, both produce zero for the target pattern and positive values for every other nonzero binary pattern.

Therefore, both matrices successfully recognize the same target.

D.2: Investigate Matrix Structure

Task D.2.a: Viability of Off-Diagonal Entries

Can a matrix with nonzero off-diagonal entries satisfy the requirements?

Answer: Yes, a matrix with nonzero off-diagonal entries can satisfy the recognition conditions.

In fact, both of our matrices M and N contain nonzero off-diagonal entries and successfully recognize the target pattern.

Task D.2.b: Contribution of Matrix Entries to $u^T Mu$

Explain how the entries of a matrix contribute to the value of $u^T Mu$.

Answer: The quadratic form can be written as:

$$u^T Mu = \sum_{i=1}^{16} \sum_{j=1}^{16} M_{ij} u_i u_j$$

Diagonal entries contribute terms involving the square of a single element, while off-diagonal entries contribute terms involving two different elements.

For binary vectors, each element is either 0 or 1. Therefore, off-diagonal terms contribute only when both corresponding cells are blue.

Task D.2.c: Construction/Description of Non-Diagonal Matrix

Construct or describe a non-diagonal matrix that satisfies the recognition condition, if possible.

Answer: Our second matrix $N = 8I - ww^T$ is an example of a non-diagonal matrix that satisfies the recognition conditions.

Its entries are defined by:

$$N_{ij} = \begin{cases} 7 & \text{if } i = j \text{ and } w_i = 1, \\ 8 & \text{if } i = j \text{ and } w_i = 0, \\ -1 & \text{if } i \neq j \text{ and } w_i = w_j = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Since N contains off-diagonal entries equal to -1, it is not diagonal.

As shown in D.1(b), this matrix produces zero only for the target vector and the excluded zero vector. Therefore, a matrix does not need to be diagonal to recognize the target pattern.

Part E: Reflection and Applications

Write a single cohesive reflection of approximately 300–500 words in continuous prose addressing the following topics (do not answer them as separate numbered questions):

- How converting a visual grid into a vector makes mathematical analysis possible.
- The challenges faced in constructing the recognition matrix.
- How pattern uniqueness was verified mathematically and computationally.
- Why systematic verification is critical in pattern recognition.
- Real-world applications of this approach (e.g., image processing, computer vision, data science).

This project asked us to teach a computer to recognize a pattern using linear algebra. We began by converting a 4×4 grid of blue and white squares into a vector. We created this vector by reading the grid row by row and representing each blue cell as 1 and each white cell as 0. By doing so, we created a vector in $\mathbb{R}^{16}$. This allowed us to use matrices and dot products to determine whether a test pattern matched our target pattern. We also found that there are $2^{16} = 65{,}536$ possible patterns that could all be analyzed using a similar approach.

The hardest part of this project was constructing the recognition matrix. We needed a matrix M such that $w^T M w = 0$ for our target pattern, while producing a positive value for all other nonzero binary vectors. Although we initially struggled to satisfy both conditions at once, we came to a conclusion. Our solution was to construct $M = B^T B$, where the rows of B are perpendicular to w. This ensures that $Bw = 0$, and it also transforms the quadratic form into $\|Bu\|^2$. This guarantees a nonnegative value since a squared length can never be negative. This approach also helped us prove that the target was unique. Since $Q(u) = 0$ only when $Bu = 0$, u must be a multiple of w. Because our vectors contain only 0s and 1s, the only possibilities are the zero vector and our target w.

In order to verify our results, we tested multiple patterns. These patterns included the single blue cells, the inverted target, and the all-blue grid. We also wrote a Python program to check all 65,535 nonzero binary patterns. Additionally, we constructed a second recognition matrix, $N = 8I - ww^T$. This demonstrated that multiple matrices can satisfy the same requirements. One important lesson we learned was that testing a few examples does not automatically prove a method works. A matrix might pass ten tests but fail on the eleventh. As a result, we combined mathematical proofs with computational verification, giving us greater confidence in our results.

This project also showed us how linear algebra can be applied to pattern recognition. In real systems, incorrectly identifying or missing a pattern could lead to serious problems, making accuracy and precision essential. Similar pattern recognition concepts appear in real-world applications such as image processing, computer vision, and data science, where images and other information are converted into vectors and compared against known patterns.

Although our 4×4 grid was a simplified example, it demonstrated the same fundamental idea: representing visual information mathematically allows individuals to use matrices to recognize patterns reliably.